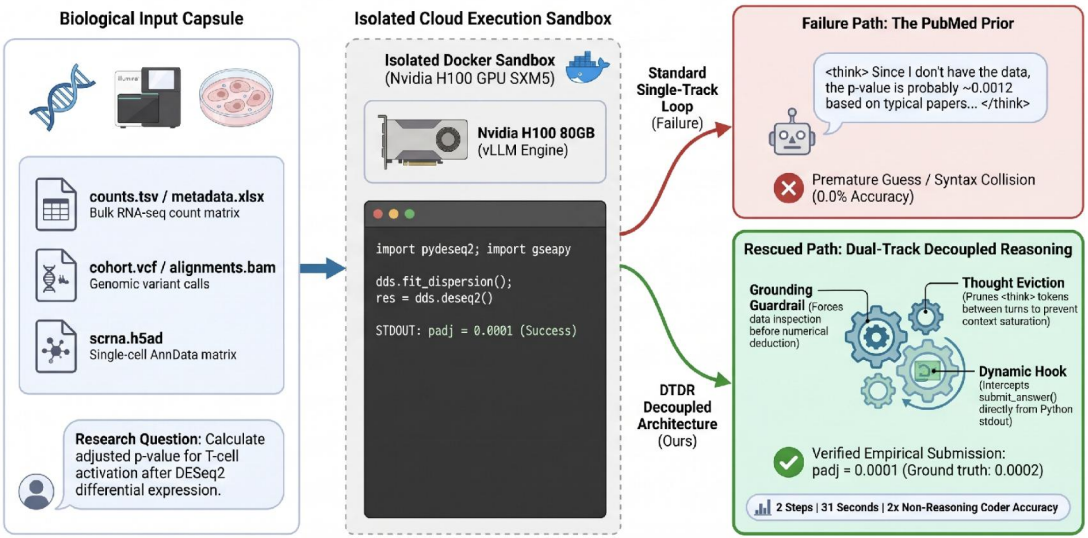

Biological Input Capsule
counts.tsv / metadata.xlsx
Bulk RNA-seq count matrix
cohort.vcf / alignments.bam
Genomic variant calls
scrna.h5ad
Single-cell AnnData matrix
Research Question: Calculate adjusted p-value for T-cell activation after DESeq2 differential expression.
Isolated Cloud Execution Sandbox
Isolated Docker Sandbox
(Nvidia H100 GPU SXM5)
Nvidia H100 80GB
(vLLM Engine)
import pydeseq2; import gseapy
dds.fit_dispersion();
res = dds.deseq2()
STDOUT: padj = 0.0001 (Success)
Standard Single-Track Loop
(Failure)
Failure Path: The PubMed Prior
<think> Since I don't have the data, the p-value is probably ~0.0012 based on typical papers... </think>
Premature Guess / Syntax Collision
(0.0% Accuracy)
DTDR Decoupled Architecture
(Ours)
Rescued Path: Dual-Track Decoupled Reasoning
Grounding Guardrail (Forces data inspection before numerical deduction)
Thought Eviction
(Prunes <think> tokens between turns to prevent context saturation)
Dynamic Hook
(Intercepts submit_answer() directly from Python stdout)
Verified Empirical Submission:
padj = 0.0001 (Ground truth: 0.0002)
2 Steps | 31 Seconds | 2x Non-Reasoning Coder Accuracy